Linear Algebra Exam 3 Study Guide

1 Section 5.2: Inner Product Spaces

- An **inner product** (denoted as $\langle \vec{u}, \vec{v} \rangle$) is literally just any operation that satifies the following axioms (let $\vec{u}$, $\vec{v}$, & $\vec{w}$ be in a vector space V, and $c \in \mathbb{R}$):
  1. $\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$
  2. $\langle \vec{u}, \vec{v} + \vec{w} \rangle = \langle \vec{u}, \vec{v} \rangle + \langle \vec{u}, \vec{w} \rangle$
  3. $c\langle \vec{u}, \vec{v} \rangle = \langle c\vec{u}, \vec{v} \rangle = \langle \vec{u}, c\vec{v} \rangle$
  4. $\langle \vec{u}, \vec{u} \rangle \geq 0$ & $\langle \vec{u}, \vec{u} \rangle = 0$ iff $\vec{u} = 0$

  Note: The dot product is an example of an inner product (it is also called the Euclidean inner product)
- Note: An inner product can also be done on matrices, and even definite integrals
- **Inner product space**: a vector space with an inner product defined
- **Length of a vector**: $||\vec{u}|| = \sqrt{\langle \vec{u}, \vec{v} \rangle}$
- **Distance between 2 vectors**: $d(\vec{u}, \vec{v}) = ||\vec{u} - \vec{v}||$
- **The angle between 2 non-zero vectors**:

$$\cos(\theta) = \frac{\langle \vec{u}, \vec{v} \rangle}{||\vec{u}||||\vec{v}||}, \text{ \& } 0 \leq \theta \leq \pi$$

- **Orthogonal**: $\langle \vec{u}, \vec{v} \rangle = 0$
- **Unit vector eq**: $\vec{u} = \frac{\vec{v}}{||\vec{v}||}$ (or to normalize)
- **Orthogonal Projections**: a vector broken down into 2 parts
  - $\vec{\mathrm{u}} = \hat{\mathrm{u}} + \vec{z}$
    - $\vec{\mathrm{u}}$: a vector
    - $\hat{\mathrm{u}}$: the projection of $\vec{\mathrm{u}}$ on another vector $\vec{v}$. Think of it has the shadow that $\vec{\mathrm{u}}$ casts on all scalar multiples of $\vec{v}$
    - $\vec{z}$: The vector orthogonal to $\vec{v}$
  - $\hat{\mathrm{u}} = \mathrm{proj}_v \vec{\mathrm{u}} = \frac{\langle \vec{\mathrm{u}}, \vec{v} \rangle}{\langle \vec{v}, \vec{v} \rangle} \vec{v}$
  - $\vec{z} = \vec{\mathrm{u}} - \hat{\mathrm{u}}$

2 Section 5.3: Orthogonal Basis - Gram-Schmidt

- Vector spaces can have many basis, like how $\mathbb{R}^3$ has the basis $B = \{(1,0,0), (0,1,0), (0,0,1)\}$ (the standard basis for $\mathbb{R}^3$). Standard basis have some important characteristics, like how all the vectors in it are mutually orthogonal

$$(1,0,0) \cdot (0,1,0) = 0$$
$$(1,0,0) \cdot (0,0,1) = 0$$
$$(0,1,0) \cdot (0,0,1) = 0$$

- **Orthogonal set**: $S = \{\vec{v}_1, \vec{v}_1, \ldots, \vec{v}_n\} \in \mathbb{R}^n$ such that $\langle \vec{u}_i, \vec{u}_j \rangle = 0 \; \forall \; i \neq j$. Basically, all vectors are orthogonal to each other
- **Othonormal set**: $S = \{\vec{v}_1, \vec{v}_1, \ldots, \vec{v}_n\} \in \mathbb{R}^n$ such that: $\langle \vec{u}_i, \vec{u}_j \rangle = 0 \; \forall \; i \neq j$, and every vector is a unit vector.

- **Gram-Schmidt Orthonormalization Algorithm**: Algorithm to find orthonormal bases, where you start with a basis, you turn it into an orthonormal basis, then normalize each vector.
  1. Let $B = \{\vec{v}_1, \vec{v}_1, \dots, \vec{v}_n\}$ be a basis for an inner product space V
  2. Let $B' = \{\vec{w}_1, \vec{w}_1, \dots, \vec{w}_n\}$, where

$$\begin{aligned}
\vec{w}_1 &= \vec{v}_1 \\
\vec{w}_2 &= \vec{v}_2 - \frac{\langle \vec{v}_2, \vec{w}_1 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle}\vec{w}_1 \\
\vec{w}_3 &= \vec{v}_3 - \frac{\langle \vec{v}_3, \vec{w}_1 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle}\vec{w}_1 - \frac{\langle \vec{v}_3, \vec{w}_1 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle}\vec{w}_1 \\
&\vdots \\
\vec{w}_n &= \vec{v}_n - \frac{\langle \vec{v}_n, \vec{w}_1 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle}\vec{w}_1 - \frac{\langle \vec{v}_n, \vec{w}_1 \rangle}{\langle \vec{w}_1, \vec{w}_1 \rangle}\vec{w}_1 - \cdots - \frac{\langle \vec{v}_n, \vec{w}_{n-1} \rangle}{\langle \vec{w}_{n-1}, \vec{w}_{n-1} \rangle}\vec{w}_{n-1}
\end{aligned}$$

     Then B' is an orthonormal basis for V
  3. Let $\vec{u}_i = \frac{\vec{w}_i}{||\vec{w}_i||}$. Then $B'' = \{\vec{u}_1, \vec{u}_1, \dots, \vec{u}_n\}$ is an orthonormal basis for V

  Also, span$\{\vec{v}_1, \vec{v}_1, \dots, \vec{v}_n\}$ = span$\{\vec{u}_1, \vec{u}_1, \dots, \vec{u}_n\}$ for $k = 1, 2, \dots, n$

3 Section 6.1: Introductions to Linear Transformations

- **Linear Transformations**: Denoted as $T : \mathbb{R}^n \to \mathbb{R}^m$, and is a function that satifies for $\vec{u}, \vec{v} \in \mathbb{R}^n$ and $k \in \mathbb{R}$ the following:
  - The additive property: $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ (or addition)
  - The homogeneity property: $T(k\vec{u}) = kT(\vec{u})$ (or scalar multiplication)
  - If there is an $A_{m\times n}$ matrix that applies the linear transformation through matrix multiplication.
    - $T(\vec{x}) = A_{m\times n}\vec{x}_{n\times 1}$
    - If $T : \mathbb{R}^n \to \mathbb{R}^n$, then $A_{n\times n}$
  - $T(-\vec{v}) = -T(\vec{v})$
  - $T(0) = 0$
- **Zero transformation**: $T(\vec{v}) = 0,$ for all $\vec{v}$
- **Identity transformation**: $T(\vec{v}) = \vec{v},$ for all $\vec{v}$

4 Section 6.2: The Kernel and Range of Linear Transformation

- **Kernel**: all vectors $\vec{v}$ in V such that that satisfy $T(\vec{v}) = 0$ where $T : V \to W$. Denoted by ker(T)
  - Looking at $T(\vec{x}) = A\vec{x}$, with the kernel would be all $\vec{x}$ where $A\vec{x} = 0$. This is also the nullspace of A
  - If $T : \mathbb{R}^n \to \mathbb{R}^m$, kernel is in $\mathbb{R}^n$
- **Range**: of a linear transformation $T : V \to W$ is a subspace of W. Think of it as the same as the range of a function, where the range is a subset of the codomain
  - Looking at $T(\vec{x}) = A\vec{x}$, the range would be the column space of A
  - If $T : \mathbb{R}^n \to \mathbb{R}^m$, range is in $\mathbb{R}^m$
- **The dimension theorem**: says that rank(T)+ kern(T) = dimension of the domain

  - rank(T) is the dimension of the row space or column space of T, or how many vectors are in the row space or column space

- **One-to-one (or bijective)**: When $T(\vec{v}_1) = T(\vec{v}_2)$ iff $\vec{v}_1 = \vec{v}_2$ (same as one-to-one in in a function)
  - A transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is one-to-one iff ker$(T) = \vec{0}_n$
    - If $R = $ rref(A) has non-pivot columns (free variables) then T is not one-to-one
- **Onto transformation**: When $T : \mathbb{R}^n \to \mathbb{R}^m$ range is all of $\mathbb{R}^m$ (same as onto in a funcion)
  - Onto iff rank$(T) = m$
  - When $R = $ rref(A) has at least one row consisting all zeros, then not onto

5 **Section 6.3**: Matrices for Linear Transformations

- Example: 2 representations of $T : \mathbb{R}^3 \to \mathbb{R}^3$
  - $T(x_1, x_2, x_3) = (2x_1 + x_2 - x_3, -x_1 + 3x_2 - 2x_3, 3x_2 + 4x_3)$
  - $T(\vec{x}) = A\vec{x} = \begin{bmatrix} 2 & 1 & -1 \\ -1 & 3 & -2 \\ 0 & 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

  The second is simpler to write and understand
- **Matrix addition/subtraction**: Given $A = \begin{bmatrix} a_{ij} \end{bmatrix}_{m\times n}, B = \begin{bmatrix} b_{ij} \end{bmatrix}_{m\times n}, A + B = \begin{bmatrix} a_{ij} + b_{ij} \end{bmatrix}$ and $1 \le i \le m$ and $1 \le j \le n$
- **Recall**: $f \circ g = f(g(x))$
  - The same can be done with transformations: $T_1 : \mathbb{R}^n \to \mathbb{R}^k, T_2 : \mathbb{R}^k \to \mathbb{R}^m, T_2 \circ T_1 : \mathbb{R}^n \to \mathbb{R}^m$

$$\vec{u} \in \mathbb{R}^n$$
$$T_1(\vec{u}) = \vec{v} \in \mathbb{R}^k$$
$$T_2(\vec{u}) = \vec{w} \in \mathbb{R}^m$$
$$(T_2 \circ T_1)(\vec{u}) = T_2(T_1(\vec{u})) = T_2(\vec{v}) = \vec{w}$$

- **Inverse Linear Transformation**: If $T_1 : \mathbb{R}^n \to \mathbb{R}^n$ and $T_2 : \mathbb{R}^n \to \mathbb{R}^n$ are linear transformations such that for every $\vec{v}$ in $\mathbb{R}^n$,
$$T_2(T_1(\vec{v})) = \vec{v}$$
  and
$$T_1(T_2(\vec{v})) = \vec{v}$$
  then T_2 is the **inverse** of T_1, and T_1 is **invertible**

6 **Section 7.1**: Eigenvalues and Eigenvectors

- Eigenvalue problem: When A is an $n \times n$ matrix, do nonzero vectors $\vec{x}$ in $\mathbb{R}^n$ exist such that $A\vec{x}$ is a scalar multiple of $\vec{x}$, or $A\vec{x} = \lambda\vec{x}$?
  - **Eigenvalue**: scalar denoted as λ
  - **Eigenvectors**: The nonzero vector $\vec{x}$ and cannot be zero
- **Eigenspaces**: The set of all eigenvectors of a given eigenvalue λ, together with the $\vec{0}$. Since each eigenvalue can have infinitely many eigenvectors. This is also a subspace of $\mathbb{R}^n$.
- **Finding eigenvalue of A**: a scalar such that det$(\lambda I - A) = 0$

- $|\lambda I - A| = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_2\lambda^2 + c_1\lambda + c_0$
- Finding the roots of the above give you the eigenvalues of A

- **Finding eigenvectors of** λ: The nonzero solutions of $(A - \lambda I)\vec{x} = 0$
  - For each λ, find the eigenvectors by solving the above equation. Requires row reducing, and there must be at least one row of zeros
- **Eigenvalues of Triangular Matrices**: If A is an $n \times n$ triangular matrix, then its eigenvalues are the entries on its main diagonal

7 Section 7.2: Diagonalization

- Diagonalization problem: For a square matrix A, does there exist an invertible matrix P such that P^-1AP is diagonal?
- **Square matrix**: Diagonalizable when A is similar to a diagonal matrix D. Or when there is an invertible matrix P such that $D = P^{-1}AP$.
  - A matrix A_n is diagonal if it has n linearly independent eigenvectors.
  - Similar matrices have the same eigenvalues
- **Steps for Diagonalizing a Square Matrix**: Let A be a square matrix
  - Find n linearly independent eigenvectors $\vec{p}_1, \vec{p}_2, \ldots, \vec{p}_3$ for A (if possible) with corresponding eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. If n linearly independent vectors don't exist, then A is not diagonalizable
  - Let P be the $n \times n$ matrix whose columns consist of these eigenvectors. That is $P = \begin{bmatrix} \vec{p}_1 & \vec{p}_2 & \ldots & \vec{p}_3 \end{bmatrix}$
  - The diagonal matrix $D = P^{-1}AP$ with have the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ on its main diagonal.

8 Section 7.3: Symmetric Matrices and Orthogonal Diagonalization

- **Symmetric matrix**: When a square matrix A is equal to its transpose, $A = A^{\mathrm{T}}$. Guaranteed to be diagonalizable
- **Symmetric matrix properities**:
  1. A is diagonalizable
  2. All eigenvalues of A are real
  3. If λ is an eigenvalue of A with multiplicity k, then λ has k linearly independent eigenvectors (or that the eigenspace of λ has dimension k)
  4. If λ_1 and λ_2 are distinct eigenvalues of A, then their corresponding eigenvectors $\vec{x}_1$ and $\vec{x}_2$ are orthogonal
  5. A is orthogonally diagonalizable (and has real eigenvalues) iff A is symmetric
- **Orthogonal matrices**: When a square matrix P is invertible, and $P^{-1} = P^{\mathrm{T}}$. Also, its column vectors form an orthonormal set
- **Orthogonal Diagonalization of a Symmetric Matrix**: Let A be an $n \times n$ symmetric matrix.
  1. Find all eigenvalues of A and determine the multiplicity of each.
  2. For each eigenvale of multiplicity 1, find a unit eigenvector. (Find any eigenvector and then normalize it.)
  3. For each eigenvalue of multiplicity of $k \geq 2$, find a set of k linearly independent eigenvectors. If this set is not othonormal, then apply the Gram-Schmidt orthonormalization process
  4. The results from 2 and 3 produce an orthonormal set of n eigenvectors. Use these to form the columns of P. The matrix $P^{-1}AP = P^TAP = D$ will be diagonal.